

*MA.8.AR.2.3***Benchmark**

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Benchmark Clarifications:

Clarification 1: Instruction focuses on understanding that when solving $x^2 = p$, there is both a positive and negative solution.

Clarification 2: Within this benchmark, the expectation is to calculate square roots of perfect squares up to 225 and cube roots of perfect cubes from -125 to 125.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.2.2
- MA.8.NSO.1.1, MA.8.NSO.1.7
- MA.8.GR.1.1, MA.8.GR.1.2

Terms from the K-12 Glossary

- Integer
- Real Numbers
- Whole Numbers

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.2.2

Next Benchmarks

- MA.912.AR.3.1

Purpose and Instructional Strategies

In previous courses, students wrote and solved one-step equations in one variable. In grade 7 accelerated, when given an equation in the form $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, students determine the real solutions. In Algebra 1, students will write and solve quadratic equations over the real number system.

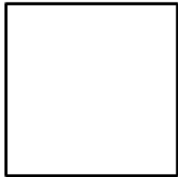
- This benchmark involves students understanding the concepts of how to square a number and find the square root as well as how to cube a number and find the cube root.
- Students should recognize that squaring a number and taking the square root of a number are inverse operations, therefore, cubing a number and taking the cube root are inverse operations as well. Students should use this understanding to solve equations containing square or cube numbers.
- In finding the square root, instruction involves discussion that there is both a positive and negative solution. Instruction can include relating the lengths of the sides of a square for square root and the length of the side of a cube to cube roots.
- Within this benchmark, it is not the expectation that students are required to isolate the x^2 term or the x^3 term when solving an equation.

Common Misconceptions or Errors

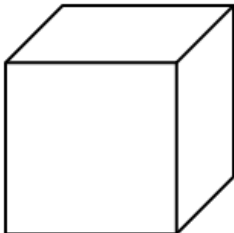
- Students may incorrectly conclude that squaring a number means to multiply by 2. Likewise, cubing may be mistaken as multiplying by 3. Use length to show doubling and area of a square to show an exponent of 2. Use of two-dimensional and three-dimensional manipulatives (*MTR.2.1*) may also help to emphasize squares and cubes versus increasing length.
- Students may think that since a negative number has no square root in the real number system, then a negative number has no cube root in the real number system.

Strategies to Support Tiered Instruction

- Instruction includes modeling the differences between doubling and squaring a value using a graphic organizer. Doubling a value would be represented by multiplying a given length by 2 whereas squaring a number would be represented by the area of a square with a given length.
 - For example, students can be given the table below to show how the left column doubles a length whereas the right column squares a length.

Doubling	Squaring
Given length of 5 _____	Given length of 5 _____
Representing $5 \cdot 2$ _____	Representing 5^2 

- Instruction includes modeling the differences between tripling or cubing a value using a graphic organizer. Tripling a value would be represented by multiplying a given length by 3, whereas cubing a number would be represented by the volume of a cube with a given length.
 - For example, students can be given the table below to show how the left column triples a length whereas the right column cubes a length.

Tripling	Cubing
Given length of 5 _____	Given length of 5 _____
Representing $5 \cdot 3$ _____ _____ _____	Representing 5^3 

- Instruction may include providing students with the opportunity to develop their own note sheet or graphic organizer for the cubes of numbers from -5 to 5.

x	x^2	x^3
-5	25	-125
-4	16	-64
-3	9	-27
-2	4	-8
-1	1	-1
0	0	0
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125

Instructional Tasks

Instructional Task 1 (MTR.7.1)

A square tile in a kitchen has an area of 121 square inches.

Part A. What is the length of one side of the square tile in inches? Is this tile smaller or larger than a one-foot by one-foot tile?

Part B. The owner of the house, Kiana, wants to put larger tiles in their kitchen to change the look of the kitchen. The new tile is a square with an area of 196 square inches.

- What is the length of the side of the new tile?
- How does this larger tile compare to the current tile used in the kitchen?

Part C. A third tile has a side length of $2\sqrt{11}$. Kiana is trying to determine which square tile covers the most area. Put the tile's side lengths of the tiles in order from greatest to least. Justify your thinking.

Instructional Task 2 (MTR.3.1)

The volume of a large cube is 125 cubic inches. The volume of a small cube is 27 cubic inches. What is the difference between the length of one side of the large cube and the length of one side of the small cube?

Instructional Task 3 (MTR.1.1, MTR.3.1, MTR.4.1, MTR.5.1, MTR.6.1)

Part A. Using the equations below, discuss with a partner the possible values of x .

- $x^2 = 64$
- $x^3 = 64$

Part B. Discuss with a partner the possible values of x if $x^3 = -64$.

Part C. With your partner, design a short presentation on the relationship between square roots and cube roots, and the possible values of the roots.

Instructional Items

Instructional Item 1

An equation is given.

$$x^2 = 49$$

What are the values of x ?

Instructional Item 2

Solve for b in the equation $-64 = b^3$.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*